Review plan · for the lecturer

Teaching databases and data engineering at university: unit check

Data engineering · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Normalisation
- Indexing
- Transactions and CAP
- Data quality

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- Which real dataset will you use?

Exit ticket

- One change to my database course this term, and how I will know it worked.

Try it in your next lecture
Bring your real-data project brief and rubric to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **R(A, B, C) with A → B and B → C, key A. The highest normal form is**

◯ BCNF ◯ 2NF ◯ 3NF

2 **Compute the closure.**

With A → B and B → C, $\{A\}^+ = \{A, B, _____\}$.

Word bank: D · ∅ · C

3 **For a filter that matches a large share of a table's rows, scanning the table can be cheaper than using an index.**

◯ True ◯ False

4 **A B+-tree with fan-out 100 over 10^6 keys has about**

◯ 3 levels ◯ 6 levels ◯ 20 levels

5 **Which can an index on (dept, year) answer with one contiguous range? Choose all that apply.**

☐ dept = 'CS' AND year = 2 ☐ year = 2 ☐ dept = 'CS'

6 **Two transactions overwrite each other's seat count. This is**

◯ A phantom ◯ A dirty read ◯ A lost update

7 **Snapshot isolation prevents write skew.**

◯ True ◯ False

8 **During a partition, a system that refuses writes to stay consistent gives up**

◯ Atomicity ◯ Availability ◯ Durability

9 **Match each problem to its dimension.**

Duplicate rows	• • • •	Validity
Grade of 140	• • • •	Completeness
Missing values	• • • •	Uniqueness

10 **Best evidence of quality in a student data project:**

◯ The number of tables ◯ The length of queries ◯ Checks, constraints and a cleaning log

Answers · for the lecturer

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1 **2NF**

B → C is transitive, so 3NF fails.

2 **With A → B and B → C, $\{A\}^+ = \{A, B, C\}$.**

Apply the dependencies until nothing changes.

3 **True**

Unselective predicates are cheaper as scans: random page reads through the index cost more than one sequential scan.

4 **3 levels**

$100^3 = 10^6$.

5 **dept = 'CS'; dept = 'CS' AND year = 2**

Only a prefix of the columns gives one range; rows with year = 2 are spread across every dept.

6 **A lost update**

One write is lost.

7 **False**

It allows it; serializable is needed.

8 **Availability**

CAP during a partition.

9 **Duplicate rows — Uniqueness**
Grade of 140 — Validity
Missing values — Completeness

Name it, then test it.

10 **Checks, constraints and a cleaning log**

Process shows trustworthiness.